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Digital product passport - Data carriers

Passeport numérique des produits - Supports de données

Digitaler Produktpass - Datenträger

This draft European Standard is submitted to CEN members for enquiry. It has been drawn up by the Technical Committee CEN/CLC/JTC 24.

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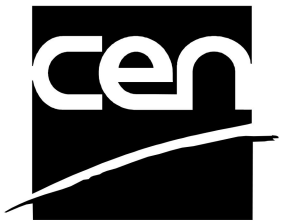

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European foreword

This document (prEN 18220:2025) has been prepared by Technical Committee CEN/CENELEC/JTC 24 Digital product passport – Framework and systems ", the secretariat of which is held by DIN.

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This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this document.

Introduction

In 2019, the European Commission introduced the European Green Deal, a detailed strategic plan aimed at sustainable growth. A key component of this plan is the Ecodesign for Sustainable Product Regulation (ESPR)[1], which is designed to guide the European Union towards achieving climate neutrality by 2050. The ESPR promotes a vision of a society that is equitable and thriving, functioning within a modern, competitive, and circular economy, all while preserving a toxin-free environment.

In detail, the ESPR mandates the adoption of production and consumption patterns that are in harmony with the Union's comprehensive sustainability goals related to climate change, environmental protection, energy use, resource efficiency, and biodiversity conservation, all within defined planetary boundaries. To achieve these goals, the regulation introduces a set of stringent ecodesign requirements.

These are specifically crafted to enhance the durability, reliability, repairability, upgradability, reusability, and recyclability of products. Such measures are vital for reducing waste, diminishing the presence of hazardous substances in products, and improving their energy and resource efficiency. Collectively, these requirements establish a rigorous framework for sustainability within the industry, pivotal for supporting the Union's transition to sustainability.

To support the implementation of these comprehensive requirements, the ESPR highlights Digital Product Passports (DPP) as a key element and a pivotal tool to ensure that all stakeholders in the value chain—manufacturers, distributors, consumers, and recyclers— have access to essential, traceable, and reliable product information, supporting informed consumer choices, and promoting better resource management and sustainability practices.

The roll-out of the digital product passport (DPP) concept is set to occur gradually over the coming years. Specific details, including what will be included in the DPP for various product groups and intermediate products, will be outlined in delegated acts. This approach ensures a comprehensive strategy that balances environmental objectives with regulatory requirements and stakeholder expectations.

To support the implementation of the DPP concept, this document addresses suitable data carriers and show how they meet the requirements. The ESPR defines 'data carrier' as a linear barcode symbol, a two-dimensional symbol or other automatic identification data capture medium that can be read by a device. It further stipulates that delegated acts will specify one or more data carriers to be used. This document specifies common rules for how to construct the automatic identification and data capture (AIDC) media to be used as data carrier linked to the product passport. These rules are based on the requirements derived from the ESPR and from the subsequent standardisation request issued by the European Commission. The requirements relate to the encoding capability, the dimensional characteristics, quality, persistence and durability as applicable. This document includes data carriers and shows how they meet the requirements.

1 Scope

This document defines requirements for data carriers used in a digital product passport system. This covers: symbology characteristics, format, error correction codes, encoding methods, printing and production quality, and durability.

This document also defines requirements on graphical or other indicators for easy recognition of DPP data carriers and the indication on the data carrier placement, machine readability, quality checking, links between physical product and digital representation.

The following aspects are out of scope: Architecture and use cases, Secure elements and any other cryptographic security features.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 15961-1:2021, *Information technology — Data protocol for radio frequency identification (RFID) for item management — Part 1: Application interface*

ISO/IEC 15961-2:2019, *Information technology — Data protocol for radio frequency identification (RFID) for item management — Part 2: Registration of RFID data constructs*

ISO/IEC 15961-3:2019, *Information technology — Data protocol for radio frequency identification (RFID) for item management — Part 3: RFID data constructs*

ISO/IEC 18046-1:2011, *Information technology — Radio frequency identification device performance test methods — Part 1: Test methods for system performance*

ISO/IEC 18046-2:2020, *Information technology — Radio frequency identification device performance test methods — Part 2: Test methods for interrogator performance*

ISO/IEC 18046-3:2020, *Information technology — Radio frequency identification device performance test methods — Part 3: Test methods for tag performance*

ISO/IEC 19762:2025, *Information technology — Automatic identification and data capture (AIDC) techniques — Vocabulary*

ISO/IEC 29158:2020, *Information technology — Automatic identification and data capture techniques — Direct Part Mark (DPM) Quality Guideline*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 19762:2025 and the terms and definitions defined in clause 3 apply.

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
additional software
additional application, program, or tool that a user will install or access separately from the default setup of most smartphones or similar devices to interact with the digital product passport

EXAMPLE Proprietary applications, plugins or extensions, dedicated apps, specialized middleware, or customized software readers.

Note 1 to entry: Standard functionalities, such as web browsers, camera-based QR readers, and universal communication protocols (e.g. DNS, HTTP, HTTPS), are not considered as additional software.

3.2
barcode
printed data carrier indicating either a reference to a linear or a 2D matrix

Note 1 to entry: It excludes RFID data carriers.

3.3
batch
subset of a model that is grouped by the economic operator based on the identical properties

3.4
consumer
individual member of the general public purchasing or using goods, property or services for private purposes

[SOURCE: [2]]

3.5
data carrier
device or medium used to store data as a relay mechanism in an automatic identification and data capture system

3.6
digital product passport
DPP
digital record of product characteristics throughout its life cycle

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability.

3.7
economic operator
manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider

3.8
item
single unit of a model

3.9
life cycle
consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal

[SOURCE: [3]]

3.10
model
version of a product of which all units share the same technical characteristics and the same model identifier

3.11
NFC
near field communication
subset of HF passive radio-frequency identification with low range

3.12
persistence
existence, and ability to be used in services outside the direct control of the issuing assigner, without a stated time limit

[SOURCE: [4]]

3.13
product
physical goods placed on the market or put into service

3.14
radio-frequency identification
RFID
identification technology that uses electromagnetic fields to automatically identify and track tags attached to objects

3.15
RAIN
radio identification
UHF passive *radio-frequency identification* (3.14)

3.16
two-dimensional symbol
code representing data in machine-readable form by a collection of polygonal or circular cells in a regular pattern which are read optically by scanning

3.17
unique identifier
identifier which is guaranteed to be unique among all identifiers used for those objects and for a specific purpose

Note 1 to entry: A unique identifier refers to unique product identifier, unique economic operator identifier and unique facility identifier.

[SOURCE: ISO 29404:2015 (Modified: Note 1 added)]

4 Concepts

4.1 DPP objective

The general objective of the DPP is to ensure that all value chain stakeholders have access to essential, traceable, and reliable product information. This is done through a link to the web or a lookup mechanism associated with the product. The identifier is represented in a data carrier that can be read automatically. In cases where the lookup mechanism through an identifier is temporarily not available, and there is sufficient space (barcode) or memory (RFID tag), additional information may be made available in the data carrier.

4.2 Basic specifications

This document specifies the basic rules applicable to the data carriers. Additional legislations determine precisely the level of identification (model, batch, item) and the data carriers that apply to a given product group.

4.3 General considerations

4.3.1 New and non-new products

The DPP will apply to different product groups available on the market. The circular economy will facilitate products to be sold, used, re-sold and re-used multiple times. In some cases, the identifier and associated data carrier will remain unchanged through the lifecycle of the product. In other cases, if a new digital product passport is needed, then a new identifier and the associated data carrier will be required. The fact that a data carrier has to remain usable after several used-reused-repair cycles has an impact on the durability of the support of the data carrier associated to the product.

4.3.2 DPP and other applications

Unique identification and associated data carrier technologies are widely used in some sectors for trade and logistic applications, consumer communication via the web and support for other regulatory requirements. Depending on the requirements set for a given product group, it may be that the existing data carriers related to the product meet the DPP requirements, at least partially. The data carrier specification for DPP should thus take into consideration the state of identification and data carriers' usage in the product group and this usage includes production specifications established by existing industry standards (eg: barcode module size, print quality and location, RFID tag frequency or memory constraints). This is to say that DPP implementation will not necessarily imply the use of a new data carrier.

4.3.3 Data carrier users and reading devices

It is expected that the DPP concept will be used by a large number of parties, where consumers are a key target. Consumers will have the ability to discover relevant information about a product, which might influence the purchasing decision. After purchase, the DPP will enable finding useful information such as user manuals, safety instructions or guidelines for recycling.

The DPP will also be used by manufacturers, distributors, resellers, recyclers, etc., for example to verify the availability and accuracy of information related to products, and to enable the reuse, repair, and recycling of products. In addition, governmental agencies, such as customs, will use the DPP to access relevant information about products.

The devices used to read the data carrier and access DPP information are smart devices and other AIDC devices (e.g: industrial barcode scanners and RFID readers in the supply chain).

5 Requirements

5.1 General

This chapter describes the requirements that data carriers should meet, as stipulated in the ESPR regulation.

A product item shall have at least one data carrier either on the product itself, on the packaging or on documentation according to this document.

5.2 Data encoding

5.2.1 Data content

The data carrier shall encode the unique identifier that allows to access the DPP. The data carrier shall comply with the rules of the unique ID to enable a smooth interoperability between systems, including the syntax when used.

5.2.2 Data syntax

The data syntax refers to the way data is represented and structured within a data carrier. Each data carrier type has a specific set of rules for encoding data, which ensures that readers and software can interpret the data correctly. These rules include the sequence of data elements, the character sets allowed, and the inclusion of start/stop characters or check digits for validation.

Data syntaxes as specified in Module 1 are in [5] or [6] or [7] or [8] or [9]

For examples of data syntax, see Annex C.

5.2.3 Character set

The unique product identifier should be presentable in a URI, using the character set specified by IETF RFC 3986 (Uniform Resource Identifier (URI): Generic Syntax).

Other data can be encoded using characters specified in [10] (Information technology — 8-bit single-byte coded graphic character sets — Part 1: Latin alphabet No. 1) or [11] (Information technology — 7-bit coded character set for information interchange.

5.3 Data carrier reading

5.3.1 Reading process for barcodes

The process of reading the DPP data carrier is usually performed by a human operator, handling one product at a time. The reading distance of the DPP data carrier will typically be between 2 and 30 centimetres. For barcodes, this distance can vary depending on several factors, such as barcode quality, lighting conditions, barcode size.

5.3.2 Reading process for NFC

NFC technologies operate in a one-to-one communication mode between a reader and a tag. The DPP data carrier reading process is thus similar to the process used when the DPP data carrier is a bar code.

5.3.3 Reading process for UHF RFID

UHF RFID (RAIN RFID) technology is designed to read multiple tags simultaneously over a distance of several meters. Different techniques can be used to read only a specific tag, such as filtering by communications protocol, adjusting the power output of the reader and the placement of the antenna, filtering by software or letting the user choose (in a list).

5.3.4 Decoding software

For end consumer usage, the unique product identifier shall be useable to access DPP public information without the need to register, download DPP-specific software, or add user credentials. For smart devices, this implies that the decoding software should ideally be natively available in the operating system. Significant research is underway to enable singulation of UHF tags by consumer smart devices. It is expected this standard will be updated as international standards are established for this use of UHF RFID.

5.4 Marking or embedding methods on product, packaging, labelling or associated document on product

5.4.1 General

There shall be at least one data carrier containing the product identifier of the DPP provided either on the product (preferred), embedded in the product (preferred), on the packaging or in the documentation.

5.4.2 Marking on product item

This method refers to the process of permanently marking or engraving information directly onto a part or product. There are several techniques that could be used to permanently identify a product throughout its lifetime.

The advantage of product marking is that the identifier stays in principle with the product throughout its complete lifetime.

5.4.3 Marking on packaging

The representation of the identifier with a data carrier is often part of the packaging of the product. This approach is used commonly for consumer goods.

Here are some of the techniques used for packaging marking:

— Laser Marking utilises lasers to engrave information onto a part's surface. It offers high precision, readability, and durability.

— Thermal Transfer which is a printing method where a thermal print head applies heat to a ribbon, transferring ink onto the substrate, usually label or film. It is commonly used for high-quality and durable printing, such as barcodes, images and texts on labels.

— Inkjet Printing can be used for product marking, especially on surfaces where other techniques may not be suitable.

If the product is re-sold, possibly refurbished, and re-used, there is a high probability that the original packaging is damaged or no longer available. The data carrier encoding the link between the physical product and its identifier is then lost.

5.4.4 Labelling

Another method is to attach a label with the data carrier on the product or on the product packaging.

Label media should be selected to endure for the full life of the product and the environment of the product.

5.4.5 Document

The DPP product identifier may be included in a document accompanying the product. The document may be physical or digital.

If the documentation is the only source of the DPP product identifier the data carrier shall be included in the documentation accompanying the product.

— Examples of physical documentation: User manual, warranty documents customer printed receipt, conformity documents…

— Examples of digital documentation: E-commerce web page, Digital customer receipt, Digital Customer account, e-mail… We speak in this case of a digital copy of the data carrier.

5.4.6 Embedded

A data carrier used for digital product passport can be embedded into the product itself. Example: textile products or tyres".

The main interest to permanently attach data carrier to the product is to ensure DPP will provide at any time of its life cycle relevant information related to the product (consumer, authorities, recyclers....) to increase circular economy.

5.5 Dimensional characteristics for barcodes

5.5.1 Error correction

Error correction code (ECC) is a crucial feature that helps 2D symbols, such as QR Codes and Data Matrix codes, remain readable even if they are partially damaged, obscured, or dirty. ECC involves adding extra data to the barcode a higher error correction level will in theory make the code more robust, but it will also make the code larger. The error correction level will not necessarily make a barcode easier to read, if the barcode print quality is poor.

5.5.2 Data carrier size

Several factors influence the size of the data carrier:

— the type of symbology;

— the X dimension that applies for the application;

— The selected error correction rate for data carrier technologies that support error correction;

— the amount of data encoded;;

— the encoding methods of characters and character sets

— the size of the product to which the dpp data carrier applies;

— the print method and the media.

Different industries and applications may have specific standards or recommendations for the X-dimension to ensure compatibility with their scanning and printing equipment. The X-dimension shall be chosen carefully to ensure that the barcode can be scanned accurately by the intended scanning environments (e.g., Point-of-Sale and consumer mobile devices). The combination of the above factors will determine the size of the data carrier. Opting for a smaller X-dimension to reduce the overall size based on the required encoded data may lead to a higher-density barcode, which can be challenging for lower-resolution scanners to read. The X-dimension also impacts printing. Printers need to have the appropriate resolution to produce clear and distinct bars or modules at the chosen X-dimension.

5.6 Data carrier quality/performance

5.6.1 General

The DPP data carrier shall reach a minimum level of quality/performance to ensure seamless usage in different environments. This section specifies the different standards that define good implementation of different data carrier technology.

Data carriers (UHF, NFC, …) should be write protected.

5.6.2 Two-dimensional symbols

The applicable standard for 2D barcode symbols is [12], Bar code symbol print quality test specification — Two-dimensional symbols. This standard defines the methodology for grading the quality of printed 2D barcodes. The grade includes a grade level, measuring aperture and the wavelength of light used for the measurement.

The minimum quality grade is typically 1,5 / 80%/ 660, where:

— 1,5 is the overall symbol quality grade;

— 80% is the measuring aperture related to the measured X dimension;

— illumination 660 is the peak response wavelength in nanometres.

ISO/IEC 29158:2020, Direct Part Mark (DPM) Quality Guideline shall be used to assess the symbol quality of direct marked parts, where the mark is applied directly to the surface of the item and the reading device is a two-dimensional imager.

Consult industry application standards, where available, for minimum print quality specifications per the intended scanning environment for the product.

5.6.3 Barcode verification

The principle is that it is important to verify the barcodes quality, using conformant verifiers. References to:

— [13], Bar code verifier conformance specification — Part 2: Two-dimensional symbols

5.6.4 HF, UHF RFID and NFC

In case the quality/performance is to be assessed, the quality/performance of RFID (UHF or NFC) shall use the following standards:

— ISO/IEC 18046-1:2011: Information technology — Radio frequency identification device performance test methods — Part 1: Test methods for system performance

— ISO/IEC 18046-2:2020: Information technology — Radio frequency identification device performance test methods — Part 2: Test methods for interrogator performance

— ISO/IEC 18046-3:2020: Information technology — Radio frequency identification device performance test methods — Part 3: Test methods for tag performance

5.7 Data carrier design

5.7.1 Data carrier placement

To facilitate the choice of more sustainable products, DPP data carriers should be displayed in a clearly visible and identifiable way such as the only visible data carrier. In the case it is not possible to have a visible data carrier, a specific marking can inform the consumer of the presence of a data carrier. They should be identifiable as the labels, containers or tags belonging to the product in question, without customers, including potential customers, having to read the brand name and model number on the labels.

5.7.2 Human readable interpretation

Human readable interpretation (HRI) is the information encoded in the DPP data carrier presented close to it in a clearly legible font, such as OCR-B specified in [14]. Different industries may have specific recommendations for HRI. Additional legislations may specify rules applicable to different product groups.

5.7.3 Signage

A 2D data carrier that links to a DPP may have a graphical marking to distinguish it from other data carriers.

When an optional graphical marking is used, it should comply with [15][16], symbol 6452 or 6452-1.

5.7.4 Accessibility

The design of the data carrier shall take into consideration the possible disabilities of consumers willing to access the DPP information.

5.8 Other considerations

5.8.1 References to recognised standards

Data carrier and quality standards used for digital product passport (DPP) applications shall be recognized by ISO and IEC. Where necessary, other specifications may be considered in the following order of preference: existing European standards, national standards, and specifications developed by industry fora. The use of additional software to enhance accessibility may also be considered.

5.8.2 Proven technologies

The data carriers selected for DPP applications shall have proven implementations on the market, both for the production of the data carrier and for the capture and processing of the encoded data.

6 Data carrier technologies

6.1 General

This chapter introduces 2D barcodes and radio-frequency technologies. It provides an overview of their technical characteristics and shows how they meet the requirements specified in chapter 5. New data carriers emerging in the future could be included in a new version of this document and be subject to the same assessment against the identified requirements.

The detailed list of requirements expressed in the ESPR is available in Annex A and Annex B.

They are summarized as follows:

1. data capacity for URI and additional data;

2. ability to use ISO/IEC 15459 conformant identifier;

3. granularity: ability to identify the product item at model, batch and item level;

4. recognised by international standards;

5. adoption by the market;

6. data carrier scannable and readable with smart devices;

7. native scanning software at the operating system level;

8. supports error correction;

9. persistence/longevity.

The selection of the data carrier shall be performed according to Annex A.

6.2 Two-dimensional barcodes

6.2.1 General

Two-dimensional (2D) barcodes store data in both the horizontal and vertical dimensions through a pattern of dots, squares, hexagons or rectangles. 2D barcodes can in theory encode up to 7.000 characters.

6.2.2 Data Matrix

Data Matrix as specified in [17] is a compact 2D barcode that uses black and white cells arranged in a square or rectangular pattern. It can store text, numeric, or binary data and is highly efficient for encoding small amounts of information. Its robust error correction makes it ideal for marking small

items like medical devices and electronics and cosmetics and also in retail since its adoption within retail scanning systems is trending rapidly.

6.2.3 QR Code

QR code (Quick Response Code) as specified in [18] is a 2D barcode that stores text, URLs, or data. It is easily scanned by smartphones or cameras, enabling quick access to information. Known for its versatility, fast readability, and error correction, it is widely used in marketing, payments, logistics, and digital interactions. QR code adoption within retail scanning systems is trending rapidly.

6.2.4 Two-dimensional barcodes characteristics

6.2.4.1 Common characteristics

The 2D barcodes considered in this document share the following common characteristics:

a) Data capacity for URI and additional data.

All 2D barcodes referred to in this document have the capacity to encode an URI and additional data.

b) Ability to use [19], [20] conformant identifier.

c) Granularity: ability to identify the product item at model, batch and item level

Several methods enable to verify the legitimacy of a 2D barcode: Verify the check digit when available; Verify that the scanned data matches the expected data; Check the barcode's data against a trusted database; Confirm the legitimacy of the item or product associated with the barcode.

d) Persistence/Longevity.

The persistence/Longevity of a 2D barcode is related to the method used to mark the product item to which the barcode is related, as described in section 5.4.

6.2.4.2 Specific characteristics

Table 1 describes the following characteristics of each linear barcode under consideration:

1. character set;

2. data capacity;

3. recognised by international standards;

4. adoption by the market;

5. native reading software on smart devices;

6. supports error correction.

Table 1 — Specific Characteristics of 2D data carriers

		Data Matrix	QR Code
Character set		[21]	[21]
Capacity	**Digits/ AN**	**3,116 numeric, 2,335 alphanumeric characters**	**numeric, 7089, 4,296 alphanumeric, 2,953 Kanji Characters**
	bits	**12,44,**	**14,81 (H) 34,36 (L)**
	bytes	**up to 1555 bytes**	**1852 bytes (H) to 4296 bytes (L)**
Encode URI		**Yes**	**Yes**
ISO/IEC 15459		**Yes**	**Yes**
Granularity		**Yes**	**Yes**
Standard		[17]	[18]
Market adoption		**Pervasive**	**Pervasive**
Native software smart devices		**Partial/Pervasive**	**Partial/Pervasive**
Error correction		**Automatic. 25% to 33% of the symbol's data capacity**	**7% to 30% of the symbol's data capacity**
Reading/scanning range		**10 to 20 cm**	**10 to 20 cm**
Bulk reading		**No**	**No**
Power supply		No	No
Security		Limited (serial, batch, data encryption, …)	Limited (serial, batch, data encryption, …)
Cost	**Static data**	Low	Low
	Dynamic data	Low to medium	Low to medium
Frequency		N/A	N/A
Radio regulation		N/A	N/A

6.3 Radio frequency data carriers

6.3.1 General

There are many different RFID technologies. In this document, and for DPP-related purposes, we only consider 3 of them:

— High Frequency (HF) passive RFID;

— Near Field Communication (NFC);

— Ultra High Frequency (UHF) passive RFID.

NOTE Technology can be considered as a subset of HF RFID because it operates at the same frequency. Its implementation in and use with smart devices is mainly described in NFC Forum Technical Specifications.

6.3.2 HF RFID

6.3.2.1 General

The different HF (High Frequency) RFID technologies are specified in: [22], [23], [24], [25], [26], [27], [28] and [29].

6.3.2.2 General overview of HF RFID

High Frequency (HF) passive RFID is a set of RFID technologies that operate at 13,56 MHz. These technologies mainly use magnetic field coupling between readers and tags to wirelessly exchange data. They are used for proximity or vicinity applications like product identification / authentication, transport ticketing system, access control, contactless payment and much more.

6.3.2.3 Data encoding

Data encoding of HF RFID tag are defined in [30], [31], [32] and [33].

6.3.2.4 Air interface protocols

Air interface protocols for HF RFID are mainly defined in [22], [23], [24], [25], [26], [27], [28], [29].

6.3.2.5 Test methods for HF/RIFD air interface compliance

[34] defines test methods which are specific to proximity cards and objects, and proximity coupling devices and proximity extended devices, defined in [22], [23], [24] and [25].

[35] defines test methods which are specific to vicinity cards and objects, and vicinity coupling devices, defined in [26], [27], [28].

[36] defines test methods for determining the conformance of radio frequency identification devices (tags and interrogators) for item management with the specifications given in [29].

6.3.3 NFC

6.3.3.1 General overview of NFC (HF)

NFC (Near Field Communication) is a wireless communication technology that enables devices to exchange data over short distances, typically within 4 centimetres. Like HF RFID, NFC operates at 13,56 MHz. NFC is designed for secure, proximity-based interactions. It supports three modes:

— reader/writer;

— card emulation; and

— peer-to-peer communication.

6.3.3.2 Air interface protocols

NFC air interface protocols are defined in [37] and [38]. Additional information can also be found in NFC Forum Analog and Digital Specifications [39](type2, type 3, type 4, type 5).

6.3.3.3 Data Encoding

Data Encoding of UHF RFID is described in [30], [31], [32] and [33]. To be natively read by smart devices, NFC Forum Technical Specifications provides ways encode data in NFC tags [39].

6.3.3.4 Test methods for NFC air interface compliance

Compliance with NFC Forum Specifications is described in the NFC Forum Certification program [39] (NFC certification release 14).

6.3.4 UHF RFID

6.3.4.1 General overview of UHF RFID

UHF passive RFID (Radio Frequency Identification) is a wireless technology used to automatically identify and track objects using radio waves. It consists of two main components: tags, which are small devices attached to items, and readers, which emit radio waves to detect and read the tags.

This technology is widely used in industries like retail, logistics, healthcare, and manufacturing for inventory management, asset tracking, access control, and more. Unlike barcodes, UHF RFID does not require line-of-sight and can read multiple tags simultaneously, making it a highly efficient tool for tracking large quantities of items in real-time. UHF passive RFID also known as RAIN RFID.

6.3.4.2 Air interface protocols

The air interface protocol for UHF RFID is defined in ,[40]. It is also defined in [41]

6.3.4.3 Data Encoding

Data Encoding of UHF RFID is described in [40]. It defines the numbering system identifier toggle bit. If the toggle bit is set to 0_2, then the application referred to as a GS1 EPC global application and the data encoding is described in GS1 Tag Data Standard [42]. If the toggle bit is set to 1_2, the application is referred to as a non-GS1 EPC global application and the encoding shall be as defined in ISO/IEC 15961-1:2021, ISO/IEC 15961-2:2019 and ISO/IEC 15961-3:2019.

6.3.4.4 Test methods for UHF RFID air interface compliance

[43] specifies test methods for determining the conformance of radio frequency identification (RFID) devices (tags and interrogators) for item management with the specifications given in . It is also described in [44].

6.3.4.5 Radio Frequency Privacy

[45] (RFID Privacy Impact Assessment Process). This EN describes when to implement a PIA and how to assess the level of risk. It also describes all kind of possible threats and proposes mitigation techniques to reduce the risk to an acceptable level. It covers all kind of RFID technologies including UHF passive and NFC.

6.3.4.6 Radio frenquency data carriers characteristics

This clause describes the characteristics of the radio data carrier technologies.

Table 2 — Typical use and specific characteristics of RFID data carriers

		UHF RFID	NFC
Character set		Depends on the application standard	UTF-8
Capacity	Digits/ AN	Depends on the application standard	32 to ~ 16000 UTF-8 characters
	bits	96 bits typical ~ up to 480 bits. With optional additional user memory of several kbits	256 bits to ~ 130 kbits
	bytes	12 to 60 bytes	32 bytes to ~16 kbytes
Encode URI		No	Yes
ISO/IEC 15459		Yes	Yes
Granularity		Yes	Yes

		UHF RFID	NFC
Standard		-	[37], [38] Note: NFC Forum also provide additional information in NFC Forum Technical Specifications[39]
Market adoption		Pervasive	Pervasive
Native software smart devices		Partial	Pervasive
Error correction		Yes	Yes
Reading/scanning range		Up to 10 meters or more	Less than 5 cm
Bulk reading		Up to 1000 tags per second	No
Power supply		Tags can be battery assisted or derive power from the reader's electromagnetic field	Tags can be battery assisted or derive power from the reader's magnetic field
Security		Yes	Yes
Cost	**Static data**	low	low
	Dynamic data	medium	low
Frequency		860-930 MHz	13.56 MHz
Radio regulation		Regulations vary by region	Widely available

Annex A
(normative)

Criteria for selection of data carriers - Mandatory criteria

Table A.1 — Mandatory requirements for selecting the data carrier

Requirement	Reference	Data carrier status
Ref ISO/IEC	**ESPR Article 12 - 6 (b)**	
The data carrier shall connect unique identifier to a DPP	**ESPR Article 10, 1 (a)**	
DPP Access: Relevant actors shall have free of charge and easy access to the digital product passport	**ESPR Article 9, 2 (f) Article 11 (b)**	
Interoperable: data carrier and supporting systems shall be interoperable between different approaches by, for example, use of international standards	**ESPR p42 Article 12 Unique ID - 6 (a), (b), (c). Article 10 a)**	
Granularity: support differentiating model/batch/ item	**ESPR Article 9, 2. (d)**	
Manufacturability: Demonstrated compatibility with mass or serial nature of product production processes	**ESPR Art 60 - (2)**	
Improve Traceability: Enables the tracing of individual items along the value chain (and entire supply chain)	**ESPR Article 9 - 3. (c)**	
No undue restrictions: The data carrier shall not result in vendor lock-in. It shall also not impose limitations on use except where such restrictions are technically necessary	**ESPR , Article 10 (d)**	
Availability: Take into account relevant existing technical solutions and standards. Compliance by end of 2027	ESPR 12 Unique ID 6 (b)	

Annex B
(informative)

List of criteria to be considered when selecting a data carrier to ensure success

Table B.1 — Criteria to be considered when selecting a data carrier to ensure success

Requirement	Reference	Data carrier status
Native Access: Access with smart devices without need to download software		Optional
Resolvable: data carrier makes it possible to access additional information		Optional - depends on other rules.
Adopted: demonstration of broad adoption of data carrier in the market		**To be considered in the solution**
Impact of large URI: data carrier or its readability are impacted by large unique identifiers, such as full-length URLs		
Persistence/Longevity: data carrier survives the stress and wear thorough the life cycle of products	ESPR article 10/1/a, and 11/b	
Parallel reading: data carriers can be read efficiently in bulk		
Data carrier support by native reader of operating system of consumer mobile devices		
Unique ID: Is the data carrier enabling the use of identifier ISO/IEC 15459 series or similar ?		
Syntax: Is the data carrier supported by a syntax used to carry ISO/IEC 15459 series or equivalent?		
Barcode Assessment [46]. NOTE 1 This will only apply to 2D barcodes.		
[47]Information technology — Radio frequency identification device performance test methods — Part 1: Test methods for system performance [48] Part 2: Test methods for interrogator performance [49]Part 3: Test methods for tag performance NOTE 2 This will only apply to RFID		
NFC data carrier quality :		

Requirement	Reference	Data carrier status
Data carrier quality [50] Cards and security devices for personal identification — Test methods — Part 6: Contactless proximity objects NOTE 4 This will only apply to HF RFID operating according to the ISO 15693 series.		
[51], [52], NFC Performance data carrier		

Annex C
(informative)

Examples of Data Syntax

C.1 Example as described in ISO/IEC 18975

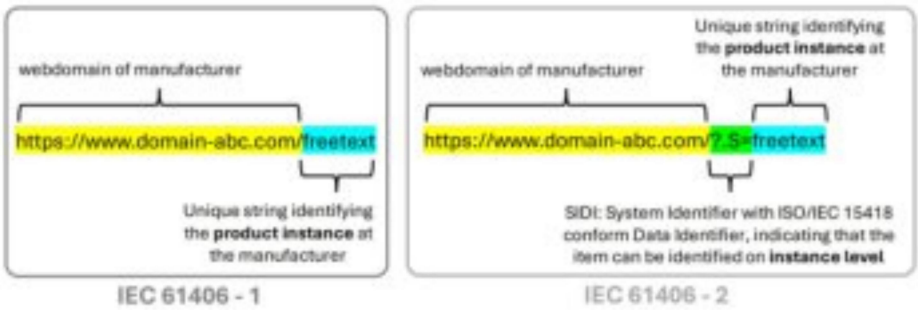
Figure C.1 provides an examples as described in [5].

Encoded data: "https://example.com/01/09524810000339/10/YA12AB?17=271231"

Figure C.1 — Example of a structured path URI that uses GS1 Application Identifiers

These ideas are expanded upon to provide a fully implementable system in GS1 Digital Link URI syntax.

C.2 Example as described in [6] and [7]



C.3 Example of [6]

Example of [6]

Encoded data: "https://www/domain-abc.com/freetext"

NOTE [18] requires a 4X quiet zone around the QR Code.

C.4 Example of [7]

Example of [7]

Encoded data: "https://www.domain-abc.com/?.P=freetext "

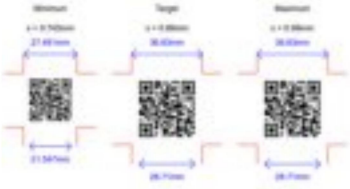
NOTE [18] requires a 4X quiet zone around the QR Code.

C.5 Examples of syntax with MH-10

These ideas are expanded upon to provide a fully implementable system in the AutoID specification.

Figure C.2 — Example of the query string approach using ANSI MH10.8.2 Data Identifiers to identify a product and its expiry date

Encoded data :"HTTPS://WWW.DOMAIN-ABC.COM?.25P=QCELMI12345&.S=654321"



C.6 Example of QR code embedding an identifier in URI a link, ISO/IEC 18975 compliant: GS1 Digital Link URI

(01)09524810000339

Annex D
(informative)

Data carriers and ID schemes for products

D.1 General

With reference to digital product passport - Unique ID Schemes, the data carriers shall be interoperable, by ensuring: *Unique Identification:* The unique product identification shall be retrievable from a data carrier as specified in EN XXXXX (no number yet), linking it to the digital product passport.

D.2 ID Scheme 5.1.2.1 Web enabled structured path identification for products

D.2.1 General

Table D.1

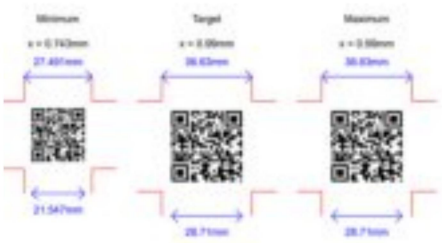
Component	Meaning	Value
GTIN (01)	**Global Trade Item Number**	**09524810000339**
Batch/Lot	**Unique item-level identifier**	**YA12AB**
Date of Expiry (17)	**YYMMDD format**	**271231**
Data carriers	**QR Code, Data Matrix and NFC examples**	
Scheme & Domain	**https://example.com**	
Encoded data	https://example.com/01/09524810000339/10/YA12AB?17=271231	

D.2.2 Example with a QR Code

Using [53] and the ISO/IEC 15459 series.

Table D.2 — QR Code estimation size (mm)

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	37 x 37	27.491	36.63	36.63
M	41 x 41	30.463	40.589	40.589
Q	45 x 45	33.435	44.55	44.55
H	49 x 49	36.407	48.51	48.51
Module x-dimensions (mm) ➔		0.743	0.99	0.99



D.2.3 Example of QR Code with [53] using GS1 Digital Link URI syntax for ECC L

Encoded data:

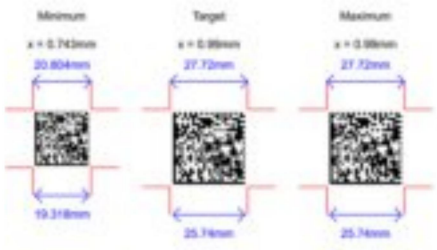
https://example.com/01/09524810000339/10/YA12AB?17=271231

D.2.4 Example with a Data Matrix

Table D.3 — Data Matrix estimation size (mm)

Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
28 x 28	20,804	27,72	27,72
Module x-dimensions (mm) ➔	0,743	0,99	0,99

D.2.4.1 Example of Data Matrix with [53] using GS1 Digital Link URI syntax



D.2.5 Example of Data Matrix with [53] using GS1 Digital Link URI syntax

Encoded data:

https://example.com/01/09524810000339/10/YA12AB?17=271231

D.2.6 Example of Data in an NFC tag

No visual representation of the encoding in an NFC tag.

D.3 ID Scheme 5.1.2.2 Web enabled query string ID for products, without structure

D.3.1 Web enabled query string ID for products

Table D.4

Component	Meaning	Value
Product ID (.25P)	IAC + CIN + Product Number	QCELMI12345
Serial number (.S)	Unique serial number	654321
Scheme and Domain	HTTPS://WWW.DOMAIN-ABC.COM	
Encoded data	HTTPS://WWW.DOMAIN-ABC.COM/?.25P=QCELMI12345&.S=654321	

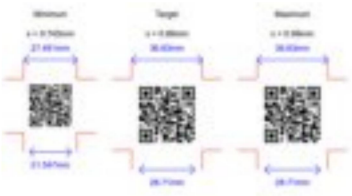
D.4 Example with a QR Code

QR Code size

Table D.5 — QR Code estimation size

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	37 x 37	27.491	36.63	36.63
M	41 x 41	30.463	40.589	40.589
Q	45 x 45	33.435	44.55	44.55
H	49 x 49	36.407	48.51	48.51
Module x-dimensions (mm) ➔		0.743	0.99	0.99

Example of web enabled query in a QR Code ECC L



D.5 ID scheme 5.2.2.1: Identification Link (IL) with structure

D.5.1 Example of Identification Link (IL) with structure, with graphical frame to indicate item

Identification Link (IL) without structure

Table D.6 — Identification link (IL) without structure

Component	Meaning	Value
Identifier (freetext)	Unstructured ID string	freetext
Scheme & Domain	https://www.domain-abc.com	
Full example	https://www.domain-abc.com/freetext	

D.5.2 QR Code size of IL with graphical marking

Table D.7

Error correction	Module Count including quiet zone	Including the graphical marking in Quiet Zone (additional 1X on each side)	Minimum size (mm)	Maximum size (mm)
L	-	-	-	-
M	37 x 37	38x38	28,23	36,63
Q	41 x 41	42x42	31,21	40,59
H	45 x 45	46x46	33,44	44,55
Module x-dimensions (mm) ➔			0,743	0,99

Table D.8 — QR Code size with the graphical Marking

Example of an Identification Link with Graphical marking

Encoded Data: "https://www.domain-abc.com/freetext" - ECC Q

NOTE [18] requires a 4X quiet zone around the QR Code.

D.6 ID scheme 5.2.2.2: Identification Link (Structured ID Link)

D.6.1 Example of Identification Link (IL) with structure and graphical frame to indicate item

Identification Link (IL) with structure

Table D.9 — Identification Link (IL) with structure and serial per [9]

Component	Meaning	Value
Scheme & Domain	Protocol and domain	https://www.domain-abc.com
Serial Number (.S)	Serial number, free format or per standard	freetext
Batch Number (.10B)	Batch or lot number	20250423
Full example:	https://www.domain-abc.com/?.S=freetext&.16D=20250423	

D.6.2 QR Code size of IL with graphical marking

Table D.10 — QR Code size with the graphical Marking

Error correction	Module Count including quiet zone	Including the graphical marking in Quiet Zone (additional 1 X on each side)	Minimum size (mm)	Maximum size (mm)
L	-	-	-	-
M	37 x 37	38x38	28,23	37,62
Q	45 x 45	46x46	34,18	45,54
H	53x53	54x54	40,12	52,47
Module x-dimensions (mm) ➔			0,743	0,99

Example of Identification Link with serial number per [9]

Encoded data: "https://www.domain-abc.com/?.S=freetext&.16D=20250423" - ECC Q

NOTE [18] requires a 4X quiet zone around the QR Code.

D.6.3 Model Identification IL (product level with optional reuse of existing ID solutions) QR Code size with graphical marking

Table D.11

Model Identification (Product Code Level with optional reuse of existing ID solutions)

Error correction	Module Count including quiet zone	Including the graphical marking in Quiet Zone (additional 2X on each side)	Minimum size (mm)	Maximum size (mm)
L	-	-	-	-
M	37 x 37	39x39	28,98	38,61
Q	45 x 45	49x49	36,41	46,53
H	53x53	57x57	42,35	56,43
Module x-dimensions (mm) ➔			0,743	0,99

Example of Model Identification :

Encoded data: " https://www.domain-abc.com/?.P=freetext" - ECC Q

NOTE [18] requires a 4X quiet zone around the QR Code.

D.6.4 Example Model Identification (Product Code Level with optional reuse of existing ID solutions)

Identification Link (IL) with structure

Table D.12 — Model Identification (Product Code Level with optional reuse of existing ID solutions)

Component	Meaning	Value
Scheme & Domain	Protocol and domain	https://www.domain-abc.com
Product Code (.P)	Model-level identifier	freetext *(or from ID scheme)*
Data carrier	QR-code example with graphical frame to indicate model.	
Full example	https://www.domain-abc.com/?.P=freetext	

D.7 ID scheme 5.3: Decentralized Identifiers

D.7.1 General

Table D.13

Component	Meaning	Value
Product DID	Full DID identifying a product (DID method = web)	did:web:abc.com:model4TR
Service Endpoint	Optional service query parameter – specifies what to retrieve	?service=item-dpp
DID Method	Method part of the DID – determines resolution rules	web
Resolution Standard	Resolver must follow Decentralized Identifier Resolution (DID Resolution) v0.3:2025	W3C-compliant
Returned Resource	Based on service: may return a UI, machine-readable Verifiable Credential, or other digital product passport data	DPP (e.g. for item)

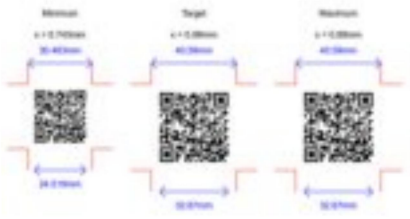
Component	Meaning	Value
Data carrier	QR-code example	
Scheme & Domain	https://resolver.io (DID resolver domain used to resolve the DID)	
Full example	https://resolver.io/did:web:abc.com:model4TR/?service=item-dpp	

D.7.2 Example with a QR Code

Using DIDs

Table D.14 — QR Code estimation size (mm)

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	41 x 41	30.463	40.589	40.589
M	41 x 41	30.463	40.589	40.589
Q	49 x 49	36.407	48.51	48.51
H	53 x 53	39.379	52.47	52.47
Module x-dimensions (mm) ➔		0.743	0.99	0.99
Example of QR Code DID ECC L				



DID in QR Code ECC L.

Contained data: "https://resolver.io/did:web:abc.com:model4TR/?service=item-dpp"

D.8 ID scheme 5.4.2.1: Product and group identification, RFID

No visual representation of the encoding in an RFID tag.

D.9 ID scheme 5.4.1.2 Product and group identification, 2D-symbols

Table D.15

Component	Meaning	Value
GTIN (01)	Global Trade Item Number used as a base for batch-level identification (ISO/IEC 15459-6)	09506000134352
Expiration date (17)	Use by or expiry	250101
Data carrier Syntax	Encoded using high-capacity AIDC media syntax (e.g., QR Code) per ISO/IEC 15434	Encoded in QR Code using ISO/IEC 15434
Resolution	Web resolver returns batch-level Digital Product Passport	Handled via example.com
Data carrier	QR-Code example	
Scheme & Domain	https://example.com	
Full example	https://example.com/01/09506000134352/17/250101	

Example of QR Code Product and Group identification

Encoded data: "https://example.com/01/09506000134352/10/250101" - ECC M

Table D.16

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	37 x 37	27.491	36.63	36.63
M	37 x 37	27.491	36.63	36.63
Q	41 x 41	30.463	40.589	40.589
H	45 x 45	33.435	44.55	44.55
Module x-dimensions (mm) ➔		0.743	0.99	0.99

Example of QR Code for product and group identification :

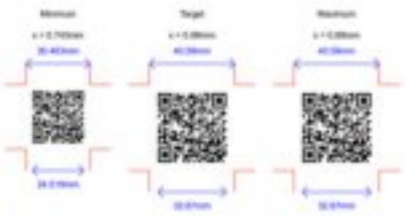
Encoded data: "https://example.com/01/09506000134352/17/250101

D.9.1 Example with a QR Code

Using DIDs

Table D.17 — QR Code estimation size (mm)

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	41 x 41	30.463	40.589	40.589
M	41 x 41	30.463	40.589	40.589
Q	49 x 49	36.407	48.51	48.51
H	53 x 53	39.379	52.47	52.47
Module x-dimensions (mm) ➔		0.743	0.99	0.99
Example of QR Code DID ECC L Encoded data: "https://resolver.io/did:web:abc.com:model4TR/?service=item-dpp"				



DID in QR Code ECC L.

Encoded data: "https://resolver.io/did:web:abc.com:model4TR/?service=item-dpp"

D.10 ID scheme 5.5: Digital Object Identifier for products

Table D.18

Component	Meaning	Value
Scheme & Domain	DOI resolver domain	https://doi.org
Path Prefix (XID)	Identifier prefix signalling ISO/IEC 15459-compliant structure	XID
DOI Indicator	Constant prefix defined in ISO 26324 for DOI	10

Component	Meaning	Value
Issued Org Code	Code assigned by the DOI issuing agency (registered under ISO/IEC 15459)	21
Product ID (Suffix)	Unique ID assigned by the org or agency (e.g., product, batch, item)	PRW82MJF
Resolution	DOI resolver translates to the digital product passport location	Handled via doi.org
Data carrier	QR-code example	
Full Identifier	ISO 26324-compliant identifier including prefix and suffix	XID10.21/PRW82MJF
Full example	https://doi.org/XID10.21/PRW82MJF	

D.10.1 Example of DOI with a QR Code

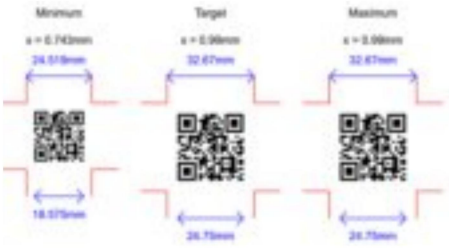
Using DOI

Table D.19 — QR Code estimation size (mm)

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	33 x 33	24.519	32.67	32.67
M	37 x 37	27.491	36.63	36.63
Q	37 x 37	27.491	36.63	36.63
H	41 x 41	30.463	40.589	40.589
Module x-dimensions (mm) ➔		0.743	0.99	0.99

Example of QR Code DOI ECC L

Encoded data: "https://doi.org/XID10.21/PRW82MJF"



Example of QR Code DOI ECC L

Encoded data: "https://doi.org/XID10.21/PRW82MJF"

Annex E
(informative)

DPP data carrier recognition

E.1 Example of Data carrier multiple use with implicit recognition

Example of QR code embedding an identifier in URI a link, [5] compliant: GS1 Digital Link URI used either for consumer engagement and price look-up at point of sales, link to DPP.

Encoded data: "https://example.com/01/09524810000339/10/YA12AB?17=271231"

E.2 Examples of data carrier with explicit recognition

An example of an explicit indication that a data carrier includes the DPP is to surround the 2D barcode by a frame and a triangularly filled corner.

This technique is specified in [6] and [7].

Example with [6]

Encoded data: "https://www/domain-abc.com/freetext"

NOTE 1 [18] requires a 4X quiet zone around the QR Code.

Example from [7]

Encoded data: "https://www/domain-abc.com/freetext?.P=feeetext"

NOTE 2 [18] requires a 4X quiet zone around the QR Code.

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of 2024/1781

This European Standard has been prepared under a Commission's standardization request.

This European Standard has been prepared under a Commission's standardization request M/604 to provide one voluntary means of conforming to essential requirements of Article 10 and 11 of 2024/1781 Ecodesign for Sustainable Products Regulation (ESPR).

Once this standard is cited in the Official Journal of the European Union under that 2024/1781, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding requirements of that 2024/1781, and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and the essential requirements of Article 9 and 10 of 2024/1781

Essential requirements specified in Article 9, 10 and 11 of 2024/1781	Clause(s)/sub-clause(s) of this EN	Remarks/Notes
Article 9, (2),N/N/ (b)	5.1	Enable to use one or more data carrier
Article (9), (c),	5.4, 5.7.1, 5.7.2	Enable data carrier presentation and positioning
Article (9), (d),	4.2, 6.1, 6.2.4	Enable multiple level of identification to be carried by the data carrier
Article (10), (1), (a)	4.3.3, 5.2.1, 5.3.1, 5.3.2, 5.4.5	Data carrier enables to access to the DPP
Article (10), (1), (b)	5.4, 5.7.1, 5.7.2	Presence of the data carrier on the product, packaging or documentation accompanying the product
Article (10), (1), (c)	5.2, 5.3	Ensures structured, machine readable, based on open standard interoperable, without vendor lock-in
Article (10), (1), (f)	4.2, 6.1, 6.2.4	Enable to provide different identification granularity
Article (10), (1), (g)	Annex A	Enable data carriers' evolutions

WARNING Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING Other Union legislation may be applicable to the falling within the scope of this standard

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